

Appendix 2

Security requirement coding scheme

Security requirement	Definition	Examples
Software independence	This code is used for cases where software independence or strong software independence is discussed or implemented as a security requirement [13, p.88]. By software independence, we refer to the requirement that if an undetected change or error occurs in a voting system's software, it should not cause a change or error in the election outcome (Ibid.).	"This system satisfies strong software independence..." "In order to achieve software independence, some states utilize permanent paper records"
End-to-end verifiability	This code is used for cases where end-to-end verifiability is discussed or implemented as a security requirement [13, p. 89]. Using Bernhard et al. definition, we define end-to-end verifiability as consisting of cast-as-intended, collected-as-cast, and counted-as-recorded verifiability (Ibid.). Note that this code should not be used if only one or two of cast-as-intended, collected-as-cast, and counted-as-recorded verifiability are mentioned. If all three apply codes would apply for a single system in the paper, code "end-to-end verifiability" instead. End-to-end verifiability should not be	"This system satisfies both individual and universal verifiability, thus ensuring end-to-end verifiability" "This system guarantees e2e verifiability"

	used for other types of verifiability. In those cases, use the appropriate “verifiability” code.	
Collection accountability	This code is used for cases where collection accountability is discussed or implemented as a security requirement [13, p.89]. If a voter can provide convincing evidence that their vote has not been collected in legitimate cases, then the system has collection accountability (Ibid.).	“It should be possible for a voter to contact the election committee in case there has been an error with their vote.”
Dispute resolution	This code is used for cases where dispute resolution is discussed or implemented as a security requirement [13, p.89]. Dispute resolution covers systems where a dispute between two participants can be resolved by a third party. This code also covers systems that are dispute free, meaning the system has built-in preventions that eliminate disputes between participants and that any third party can see if a participant has cheated (Ibid.).	“Given that our system utilizes a ballot confirmation check that returns a valid proof if the vote is valid, it can help with dispute resolutions between multiple parties.”
Verifiability • Verifiability (cast-as-intended) • Verifiability (recorded-as-cast)	This code is used for cases where verifiability and more specific versions of verifiability are discussed or implemented as a security requirement [13, p.89]. When the case mentions a specific type of verifiability, then use the appropriate subcode. If a case is verifiability, but	“Our system ensures individual verifiability” - code both “verifiability (cast-as-intended)” and “verifiability (recorded-as-cast)” since individual verifiability is usually defined as a combination of cast-as-intended and recorded-as-cast verifiability, i.e. end-to-end verifiability

• Verifiability (counted-as-recorded) • Verifiability (other)	no appropriate subcode is mentioned, use verifiability (other). If the case mentions end-to-end verifiability, use the “end-to-end verifiability” code instead. verifiability (cast-as-intended) - use for cases where cast-as-intended verifiability is discussed or implemented as a security requirement. A system has cast-as-intended verifiability if voters can verify that their cast ballot is correctly recorded (Ibid.). verifiability (recorded-as-cast) - use for cases where recorded-as-cast verifiability is discussed or implemented as a security requirement. A system has recorded-as-cast verifiability if voters can verify that their vote is collected properly in the tally (Ibid.). verifiability (counted-as-recorded) - use for cases where counted-as-recorded verifiability is discussed or implemented as a security requirement. A system has counted-as-recorded verifiability if anyone can verify that every correctly cast and collected vote is included in the tally (Ibid.). verifiability (other) - use for cases where verifiability is discussed or implemented as a security requirement, but it is not	without counted-as-recorded verifiability (King-Hang et al., 2017, p. 34). “Our system guarantees cast-as-intended verifiability.” - code “verifiability (cast-as-intended)”. “Our system guarantees recorded-as-cast verifiability.” - code “verifiability (recorded-as-cast)”. “Our system ensures universal verifiability” - code “verifiability (counted-as-recorded)” since universal verifiability is also used to describe counted-as-recorded verifiability (King-Hang et al., 2017, p. 34). “Our system ensures counted-as-recorded verifiability” - code “verifiability (counted-as-recorded)”. “Using this technique we ensure that our system has verifiability” - code “verifiability (other)” since the exact type of verifiability is not specified.
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	captured by any of the other subcodes. For example, when it is not clear what type of verifiability is discussed.	
Everlasting privacy	This code is used for cases where everlasting privacy or long-lasting privacy is discussed or implemented as a security requirement [13, p. 93]. Cases that describe systems that do not rely on cryptographic hardness to protect privacy fall under this category (Ibid.). Do not use this code when the security requirement is about weaker notions of privacy, in that case use the code "privacy".	"Everlasting privacy is ensured in our system using..." "This scheme ensures long-lasting privacy in relation to quantum adversaries"
Receipt-freeness	This code is used for cases where receipt-freeness is discussed or implemented as a security requirement [13, p.94]. A system is receipt free if a voter cannot prove how they wanted, even if they wanted to (Ibid.).	"Our system ensures receipt-freeness".
Coercion resistance	This code is used for cases where coercion resistance or uncoercionability is discussed or implemented as a security requirement [13, pp.94-95]. We define a system as coercion resistant if, given any strategy a coercer may employ, there is a way for a voter to vote in a way that is not identifiable to a coercer, enabling them to still cast their vote (Ibid.).	"Using this technique we can ensure coercion-resistance"

Availability	This code is used for cases where availability is discussed or implemented as a security requirement [13, pp.95-96]. A hybrid or fully electronic voting system satisfies availability if it can resist attempts to disrupt the election process; both partial disruptions like selective denial of service (DoS) attacks as well as complete disruptions (Ibid.).	“In order to ensure availability, we have developed a decentralized system...”
Usability • Usability (efficiency) • Usability (effectiveness) • Usability (satisfaction) • Usability (transparency) • Usability (other)	This category of codes are used for cases where usability is discussed or implemented as a security requirement [13, pp.96-97]. Usability can be discussed as its own security requirement or in a more specific form like efficiency, effectiveness, satisfaction, and transparency. Below are subcodes to classify what type of usability is discussed. usability (efficiency) - use for cases where efficiency is argued as important to security and a requirement of the system. If the case does not relate efficiency to security, then do not code it. usability (effectiveness) - use for cases where effectiveness is argued as important to security and a requirement of the system. If the case does not relate effectiveness to security, then do not code it.	“Transparency is important to ensure correct use of the system” - code “usability (transparency)”. “Usability is a core requirement in our system, the system needs to be usable to ensure...”

	usability (satisfaction) - use for cases where satisfaction is argued as important to security and a requirement of the system. If the case does not relate satisfaction to security, then do not code it. usability (transparency) - use for cases where information about the system needs to be transparent and accessible to users. usability (other) - use for cases where usability is mentioned or when there is a security requirement that relates directly to usability, but there is no appropriate subcode to describe it. If the case does not relate usability to security, then do not code it.	
Auditability	This code is used for cases where auditing is discussed or implemented as a security requirement [13, p.98].	“Auditing of the vote is a fixed step in our system, it is important to confirm the validity of the election results.”
Correctness	This code is used for cases where correctness is discussed or implemented as a security requirement (Wang et al., 2017, p. 33). Correctness in this case refers to all votes being correctly counted, meaning all valid votes are counted, and all invalid votes are not counted (Ibid.). Note that general mentions of correctness,	“... this property is needed to ensure correctness of the system”.

	such as correctness of a proof should not be coded as correctness.	
Privacy	This code is used for cases where privacy is discussed or implemented as a security requirement (Wang et al., 2017, p. 33). A system ensures privacy if the voter's ballot choices are only known to themselves (Ibid.). Systems that ensure privacy via cryptographic hardness assumptions are included. If a system does not rely on such assumptions to ensure privacy code "everlasting privacy" instead.	"... to ensure ballot secrecy..." "This would give immediate privacy..." "This ensures confidentiality of the data, only the voter can see their results" "This ensures that our system has privacy."
Unreuseability	This code is used for cases where unreuseability, the property that no voter can cast a vote twice, is discussed or implemented as a security requirement (Wang et al., 2017, p. 33).	"Our system ensures unreuseability." "No voter can cast their vote more than once."
Eligibility	This code is used for cases where eligibility, the property that only authorized voters can vote, is discussed or implemented as a security requirement (Wang et al., 2017, p. 33). We also include cases where eligibility is made verifiable for the individual or where independent parties can verify all voters are eligible.	"Our system ensures eligibility via..." "Only authorized voters are permitted to cast a vote."

Robustness	This code is used for cases where robustness is discussed or implemented as a security requirement (Wang et al., 2017, p. 33). A system has robustness if it can properly function even with some number of misbehaved voters or partial system failure (Ibid.). We do not assess whether a system satisfies a specific threshold for misbehaved voters or system failure, instead assess if the system is robust by the standards used in the specific case. For example, if the author(s) in a paper argues the system satisfies robustness, we apply the code.	"Our system ensures robustness." "The system remains functional even in the case of partial system failure or misbehaving voters."
Fairness	This code is used for cases where fairness is discussed or implemented as a security requirement (Wang et al., 2017, p. 33). This code should only be used in cases where the property of only counting the vote once all votes have been corrected such that no election authority can compute partial election results before the end of the election (Ibid.). General mentions of fairness, if not related to this requirement should not be coded.	"The system ensures fairness properties..." "The election authority cannot compute partial results before the election has ended."

Practicality	This code is used for cases where the requirement that the system should not rely on difficult to realize assumptions in large-scale scenarios, is discussed or implemented (Wang et al., 2017, p. 34).	"It is important that the system can be used for smaller elections." "... that scheme used difficult to realize assumptions whereas we have designed a protocol with weaker assumptions that can be more realistically used in live systems..."
Safe aggregation	This code is used for cases where there is a requirement of aggregating votes from electronic and non-electronic channels and to safely compute them in hybrid elections, is discussed or implemented (Cucurull et al., 2019, p. 302).	"Our system ensures safe aggregation." "Votes cast through electronic and non-electronic channels are securely combined into a single tally."
Authentication	This code is used for cases where authentication is discussed or implemented as a security requirement (Cucurull et al., 2019, p. 302). A system provides authentication if it can uniquely identify voters, and if it only authorizes eligible voters to access the vote (Ibid.).	"To ensure authentication, our system uses..." "Only authorized voters are permitted to access the ballot..." "Voters are uniquely identified via their unique ID, ensuring authentication..."
Integrity	This code is used for cases where integrity is discussed or implemented as a security requirement (Cucurull et al., 2019, p. 303). Integrity here is understood as the voter, and	"Our system ensures integrity by..."

	more broadly the election result, not being affected by the voting system or any other actor with the intent to unduly sway the result (Ibid.).	
Forward secrecy	This code used for cases where forward secrecy is discussed or implemented as a security requirement (Haines and Xavier, 2016, p.2).	"Our system ensures forward secrecy." "Past votes remain protected even if cryptographic keys are compromised in the future."
Non-valid voting capability	This code is used for cases where the requirement to allow voters to send invalid votes without compromising the integrity of the election is discussed or implemented as a security requirement (Hajian Berenjestanaki, 2024, p. 4).	"Voters can submit invalid votes without compromising the integrity of the election."
Anonymity	This code is used for cases where anonymity is discussed or implemented as a security requirement (Cucurull et al., 2019, p. 303). A system ensures anonymity if it is not possible to link an unsealed vote to a voter (Ibid.).	"Our system ensures anonymity." "It is not possible to link a vote back to the voter who cast it."
Revocation	This code is used for cases where revocation is discussed or implemented as a security requirement (Haines & Boyen, 2016, pp. 6-7). A system allows	"Our system supports revocation."

	for revocation if the voter can cancel or override their previously cast vote during the election period (Haines & Boyen, 2016, pp. 6–7).	"Voters can cancel or override their previously cast vote during the election period."
Legal requirement	This code is used for cases where legal requirements are considered, specifically those related to security. This includes cases where the publication analyzes applicable law, state it as a requirement for the voting system(s) in question, or when it is simply acknowledged that local law can and/or will affect security requirements in an e-voting system.	"Local law mandates that all cast votes must remain private..." "Law requires that all DREs have an attached printer so that a paper audit trail can be done at all times" "It is important to consider the legal context the system has to operate in as it affects how e.g. verification has to be carried out."